

THOLKAPPIAN MURUGESAN

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EDUCATIONS

Indian Institute of Information Technology (IIIT) Sri City | Chittoor

2024 - 2028

B.Tech in Computer Science and Engineering

CGPA : 8.76

Relevant Coursework: Data Structures, Operating System, Object Oriented Programming, Full Stack Architecture

WORK EXPERIENCE

Full Stack Intern | Internship

May 2026 - Jul 2026

ForgeYourPeak

Remote • Chennai

- **Architected** a custom EdTech platform utilizing ASP.NET Core and React, **replacing** legacy third-party infrastructure and **eliminating** 100% of platform transaction fees.
- **Designed** a normalized PostgreSQL database schema with Entity Framework Core, **enforcing** strict JWT Claims-Based Authorization to securely manage data for active students.
- **Prototyped** an adaptive-bitrate HLS video streaming pipeline using Cloudflare R2, **aiming to cut** bandwidth egress costs by 100% and **improve** global video load times by up to 40%.

Technologies / Skills Used : React, PostgreSQL, Tailwind CSS, .NET 9, CloudFlare R2

PROJECTS

OSS Dependency Analyser | 🌐 Github

Jul 2026 - Jul 2026

- **Engineered a Full-Stack OSS Dependency Analyzer** – .NET 9 backend with React/TypeScript frontend to scan GitHub repositories for vulnerable libraries in dependency files (package.json, pom.xml, etc.).
- **Implemented Asynchronous Processing** – Integrated Hangfire and Redis for non-blocking background jobs fetching and parsing dependencies from the GitHub API.
- **Built Real-Time Vulnerability Pipeline** – Automated cross-referencing of dependencies against NVD, calculated risk scores, and pushed updates to the frontend via SignalR.
- **Designed Interactive Dashboard** – React frontend with Recharts visualizing security metrics and vulnerability trends.

Technologies / Tools Used : .NET 9, EF Core, Hangfire, SignalR, React, PostgreSQL, Redis

ProMap | 🌐 Website 🌐 Github

- **Engineered a secure RESTful API** with 15+ endpoints using **Express and JWT/bcrypt** for secure user authentication and routing.
- **Built a responsive React frontend** utilizing **React Context API** for global state management, seamlessly handling **2 core user flows** like board creation and task assignment.
- **Designed a scalable MongoDB database** with **4 optimized schemas** (Users, Boards, Tasks, etc.), enabling efficient data retrieval and one-to-many relationship mapping.

Technologies / Tools Used : React, Node.js, Express, MongoDB

Flow | 🌐 Website 🌐 Github

- **Engineered a 3D production tracking platform** utilizing a **Vite-powered React** frontend and a **Node.js/Express backend** to orchestrate complex asset lifecycles sequentially across multiple rendering stages.
- **Implemented robust Role-Based Access Control (RBAC)** and **JWT authentication middleware**, securely isolating API routes and operational privileges among platform owners, administrators, and team members.
- **Designed a responsive asset management UI** featuring dynamic state tracking and automated deadline highlighting, utilizing **Axios** to ensure optimized, real-time client-server data synchronization.

Technologies / Tools Used : React, Node.js, Express, Axios, Mongo

SKILLS

Databases :	MongoDB, PostgreSQL
Frameworks & Libraries :	ASP.Net, Node.js, Express.js, React.js
Programming Languages :	C, C++, Java, Python, JavaScript, TypeScript, HTML, CSS
Tools & Platforms :	Git, Vercel, Render, Postman, Docker

AWARDS & ACHIEVEMENTS

- **1st Place at Innoventures** (Intracollege Hackathon)